

Follow today's class instructions (dbt only. No need to run it as a DAG)

1. Create a dbt project with snowflake connector (1 pt)

dbt init HW4

```
(data226_dbt) C:\Users\kevin\OneDrive\桌面\DATA226>dbt init HW4
07:36:49 Running with dbt=1.12.5
07:36:49 Setting up your profile.
Which database would you like to use?
[1] snowflake

(Don't see the one you want? https://docs.getdbt.com/docs/available-adapters)

Enter a number: 1
account (https://<this_value>.snowflakecomputing.com): IKFRRDP-BK44433
user (dev username): CHINGTU
[1] password
[2] keypair
[3] sso
[4] workload_identity
Desired authentication type option (enter a number): 2
private_key_path (path/to/private.key): C:\Users\kevin\OneDrive\桌面\DATA226\sjsu-data226-FA26\keys\rsa_key.p8
private_key_passphrase (passphrase for the private key, if key is encrypted):
role (dev role): ACCOUNTADMIN
warehouse (warehouse name): CTDATA226
database (default database that dbt will build objects in): HW4
schema (default schema that dbt will build objects in): raw
threads (1 or more) [1]: 1
07:37:57 Profile HW4 written to C:\Users\kevin\.dbt\profiles.yml using target's profile_template.yml and your supplied values.
07:37:57 Running dbt debug to validate the project...
07:37:57 dbt version: 1.12.5
07:37:57 python version: 3.12.14
07:37:57 python path: C:\Users\kevin\anaconda3\envs\data226_dbt\python.exe
07:37:57 os info: Windows-11-10.0.26200-SP0
07:37:57 Using profiles dir at C:\Users\kevin\.dbt\profiles.yml
07:37:57 Using dbt_project.yml file at C:\Users\kevin\OneDrive\桌面\DATA226\HW4\dbt_project.yml
07:37:57 adapter type: snowflake
07:37:57 adapter version: 1.12.1
07:37:57 Configuration:
07:37:57   profiles.yml file [OK found and valid]
07:37:57   dbt_project.yml file [OK found and valid]
07:37:57 Required dependencies:
07:37:57   - git [OK found]

07:37:57 Connection:
07:37:57   account: IKFRRDP-BK44433
07:37:57   user: CHINGTU
07:37:57   database: HW4
07:37:57   warehouse: CTDATA226
07:37:57   role: ACCOUNTADMIN
07:37:57   schema: raw
07:37:57   authenticator: None
07:37:57   oauth_client_id: None
07:37:57   query_tag: None
07:37:57   client_session_keep_alive: False
07:37:57   host: None
07:37:57   port: None
07:37:57   proxy_host: None
07:37:57   proxy_port: None
07:37:57   protocol: None
07:37:57   connect_retries: 1
07:37:57   connect_timeout: None
07:37:57   retry_on_database_errors: False
07:37:57   retry_all: False
07:37:57   insecure_mode: False
07:37:57   reuse_connections: True
07:37:57   s3_stage_vpce_dns_name: None
07:37:57   workload_identity_provider: None
07:37:57   workload_identityentra_resource: None
07:37:57   platform_detection_timeout_seconds: 0.0
07:37:57 Registered adapter: snowflake=1.12.1
07:37:58 Connection test: [OK connection ok]

07:37:58 All checks passed!
07:37:58 Your new dbt project "HW4" was created!
```

dbt debug

```
(data226_dbt) C:\Users\kevin\OneDrive\桌面\DATA226\HW4>dbt debug
07:38:28 Running with dbt=1.12.5
07:38:28 dbt version: 1.12.5
07:38:28 python version: 3.12.14
07:38:28 python path: C:\Users\kevin\anaconda3\envs\data226_dbt\python.exe
07:38:28 os info: Windows-11-10.0.26200-SP0
07:38:29 Using profiles dir at C:\Users\kevin\.dbt\profiles.yml
07:38:29 Using dbt_project.yml file at C:\Users\kevin\OneDrive\桌面\DATA226\HW4\dbt_project.yml
07:38:29 adapter type: snowflake
07:38:29 adapter version: 1.12.1
07:38:29 Configuration:
07:38:29   profiles.yml file [OK found and valid]
07:38:29   dbt_project.yml file [OK found and valid]
07:38:29 Required dependencies:
07:38:29   - git [OK found]

07:38:29 Connection:
07:38:29   account: IKFRRDP-BK44433
07:38:29   user: CHINGTU
07:38:29   database: HW4
07:38:29   warehouse: CTDATA226
07:38:29   role: ACCOUNTADMIN
07:38:29   schema: raw
07:38:29   authenticator: None
07:38:29   oauth_client_id: None
07:38:29   query_tag: None
07:38:29   client_session_keep_alive: False
07:38:29   host: None
07:38:29   port: None
07:38:29   proxy_host: None
07:38:29   proxy_port: None
07:38:29   protocol: None
07:38:29   connect_retries: 1
07:38:29   connect_timeout: None
07:38:29   retry_on_database_errors: False
07:38:29   retry_all: False
07:38:29   insecure_mode: False
07:38:29   reuse_connections: True
07:38:29   s3_stage_vpce_dns_name: None
07:38:29   workload_identity_provider: None
07:38:29   workload_identityentra_resource: None
07:38:29   platform_detection_timeout_seconds: 0.0
07:38:29 Registered adapter: snowflake=1.12.1
07:38:48 Connection test: [OK connection ok]
```

2. Set up transform models (2pt)

Dummy data build using snowflake

The screenshot shows the Snowflake SQL editor with two SQL scripts. The first script creates the `RAW.USER_SESSION_CHANNEL` table and inserts 8 rows of dummy data. The second script creates the `RAW.SESSION_TIMESTAMP` table and inserts 8 rows of dummy data.

```
USE ROLE ACCOUNTADMIN;
USE DATABASE HW4;
CREATE SCHEMA IF NOT EXISTS RAW;

CREATE OR REPLACE TABLE RAW.USER_SESSION_CHANNEL(
  userid INT,
  sessionid VARCHAR(32) PRIMARY KEY,
  channel VARCHAR(32)
);

INSERT INTO RAW.USER_SESSION_CHANNEL (userid, sessionid, channel) VALUES
(1, 's1', 'facebook'),
(2, 's2', 'google'),
(3, 's3', 'facebook'),
(4, 's4', 'youtube'),
(5, 's5', 'google'),
(6, 's6', 'instagram'),
(7, 's7', 'facebook'),
(8, 's8', 'youtube');
```

```
CREATE OR REPLACE TABLE RAW.SESSION_TIMESTAMP(
  sessionid VARCHAR(32) PRIMARY KEY,
  ts TIMESTAMP
);

INSERT INTO RAW.SESSION_TIMESTAMP(sessionid, ts) VALUES
('s1', '2026-09-01 10:00:00'),
('s2', '2026-09-01 11:15:00'),
('s3', '2026-09-02 09:30:00'),
('s4', '2026-09-02 14:45:00'),
('s5', '2026-09-03 08:20:00'),
('s6', '2026-09-03 16:10:00'),
('s7', '2026-09-04 12:00:00'),
('s8', '2026-09-04 18:30:00');
```

Results (just now)

#	number of rows inserted
1	8

Results (just now)

#	number of rows inserted
1	8

Results (just now)

#	USERID	SESSIONID	CHANNEL
1	1	s1	facebook
2	2	s2	google
3	3	s3	facebook
4	4	s4	youtube
5	5	s5	google
6	6	s6	instagram
7	7	s7	facebook
8	8	s8	youtube

Results (just now)

#	SESSIONID	TS
1	s1	2026-09-01 10:00:00.000
2	s2	2026-09-01 11:15:00.000
3	s3	2026-09-02 09:30:00.000
4	s4	2026-09-02 14:45:00.000
5	s5	2026-09-03 08:20:00.000
6	s6	2026-09-03 16:10:00.000
7	s7	2026-09-04 12:00:00.000
8	s8	2026-09-04 18:30:00.000

Making folder of transform and analytics

```
(data226_dbt) C:\Users\kevin\OneDrive\桌面\DATA226\HW4>ls
README.md  analyses  dbt_project.yml  logs  macros  models  seeds  snapshots  tests

(data226_dbt) C:\Users\kevin\OneDrive\桌面\DATA226\HW4>mkdir models\transform

(data226_dbt) C:\Users\kevin\OneDrive\桌面\DATA226\HW4>mkdir models\analytics
```

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名稱	狀態	修改日期	類型	大小
analytics	✓	2026/9/18 上午 12:55	檔案資料夾	
transform	✓	2026/9/18 上午 12:54	檔案資料夾	

1. Input (transform) tables should be built as **CTE** (involves changes to dbt_project.yml)
2. Show contents of models/transform/user_session_channel.sql and transform/session_timestamp.sql

models\source.yml and dbt_project.yml

```
version: 2

sources:
  - name: raw
    database: HW4
    schema: raw
    description: "Schema which stores raw tables (ETL)"
    tables:
      - name: user_session_channel
        identifier: user_session_channel
        description: "Session data including userId, sessionId, and channel"
      - name: session_timestamp
        identifier: session_timestamp
        description: "Session data including timestamp"

models:
  HW4:
    # Config indicated by + and
    transform:
      +materialized: ephemeral
    analytics:
      +materialized: table
```

models\transform\session_timestamp.sql

```
with source as (
  select
    userId,
    sessionId,
    channel
  from {{ source('raw', 'user_session_channel') }}
)

select *
from source
WHERE sessionId IS NOT NULL
```

models\transform\user_session_channel.sql

```
with source as (
  select sessionId, ts
  from {{ source('raw', 'session_timestamp') }}
)

select * from source
WHERE sessionId IS NOT NULL
```

dbt run (Note: I set ephemeral to transform models to output CTE instead of view)

```
(data226-dbt) C:\Users\Kevin\OneDrive\桌面\DATA226\HW4>dbt run
21:47:25 Running with dbt=1.12.5
21:47:26 Registered adapter: snowflake=1.12.1
21:47:26 Unable to do partial parsing because saved manifest not found. Starting full parse.
21:47:27 [WARNING]: Did not find matching node for patch with name 'session_summary' in the 'models' section of file 'models/schema.yml'
21:47:27 [WARNING]: Test 'test.HW4.unique_session_summary_sessionId.9c5e5661a0' (models/schema.yml) depends on a node named 'session_summary' in package '' which was not found
21:47:27 [WARNING]: Test 'test.HW4.not_null_session_summary_sessionId.e6aecla6c6' (models/schema.yml) depends on a node named 'session_summary' in package '' which was not found
21:47:27 [WARNING]: Configuration paths exist in your dbt_project.yml file which do not apply to any resources.
There are 1 unused configuration paths:
- models.HW4.analytics
21:47:27 Found 2 models, 2 sources, 562 macros
21:47:27 Concurrency: 1 threads (target='dev')
21:47:27
21:47:27 Finished running in 0 hours 0 minutes and 0.02 seconds (0.02s).
```

3. Set up analytics models (1pt)

1. Show contents of models/analytics/session_summary.sql

models\analytics\session_summary.sql

```
with user_session as (  
    select * from {{ref('user_session_channel')}}  
),  
  
session_ts as (  
    select * from {{ref('session_timestamp')}}  
),  
  
final as (  
    select  
        u.userId,  
        u.sessionId,  
        u.channel,  
        s.ts  
    from user_session u  
    join session_ts s on u.sessionId = s.sessionId  
)  
select * from final
```

dbt run

```
(data226_dbt) C:\Users\kevin\OneDrive\桌面\DATA226\HW4>dbt run  
22:08:34 Running with dbt=1.12.5  
22:08:35 Registered adapter: snowflake=1.12.1  
22:08:35 Found 3 models, 2 sources, 562 macros  
22:08:35 Concurrency: 1 threads (target='dev')  
22:08:35 1 of 1 START sql table model raw.session_summary ..... [RUN]  
22:08:40 1 of 1 OK created sql table model raw.session_summary ..... [SUCCESS 8 in 2.56s]  
22:08:46 Finished running 1 table model in 0 hours 0 minutes and 10.19 seconds (10.19s).  
22:08:46 Completed successfully  
22:08:46 Done. PASS=1 WARN=0 ERROR=0 SKIP=0 NO-OP=0 REUSED=0 TOTAL=1
```

Result table on snowflake

SELECT * FROM RAW.SESSION_SUMMARY;				
results (8 minutes ago)				
	USERID	SESSIONID	CHANNEL	TS
1	1	s1	facebook	2026-09-01 10:00:00.000
2	2	s2	google	2026-09-01 11:15:00.000
3	3	s3	facebook	2026-09-02 09:30:00.000
4	4	s4	youtube	2026-09-02 14:45:00.000
5	5	s5	google	2026-09-03 08:20:00.000
6	6	s6	instagram	2026-09-03 16:10:00.000
7	7	s7	facebook	2026-09-04 12:00:00.000
8	8	s8	youtube	2026-09-04 18:30:00.000

4. Add Snapshot to the analytics table (session_summary) (2pt)

1. Show contents of snapshots/snapshot_session_summary.sql

snapshots\snapshots_session_summary.sql

```
{% snapshot snapshot_session_summary %}

{{
  config(
    target_database='HW4',
    target_schema='analytics',
    unique_key='sessionId',
    strategy='timestamp',
    updated_at='ts'
  )
}}

select *
from {{ ref('session_summary') }}

{% endsnapshot %}
```

dbt snapshots

```
(data226_dbt) C:\Users\kevin\OneDrive\桌面\DATA226\HW4>dbt snapshot
22:19:35 Running with dbt=1.12.5
22:19:36 Registered adapter: snowflake=1.12.1
22:19:37 Found 3 models, 1 snapshot, 2 sources, 562 macros
22:19:37 Concurrency: 1 threads (target='dev')
22:19:37 1 of 1 START snapshot analytics.snapshot_session_summary ..... [RUN]
22:19:42 1 of 1 OK snapshot analytics.snapshot_session_summary ..... [SUCCESS 8 in 2.49s]
22:19:43 Finished running 1 snapshot in 0 hours 0 minutes and 6.56 seconds (6.56s).
22:19:44 Completed successfully
22:19:44 Done. PASS=1 WARN=0 ERROR=0 SKIP=0 NO-OP=0 REUSED=0 TOTAL=1
```

Result table on snowflake

SELECT FROM HW4.ANALYTICS.SNAPSHOT_SESSION_SUMMARY;									
Results (2 minutes ago)									
ID	USERID	SESSIONID	CHANNEL	TS	DBT_SQL_ID	DBT_UPDATED_AT	DBT_VALID_FROM	DBT_VALID_TO	
1	1	s1	facebook	2026-09-01 10:00:00.000	9c115f67d200617b8f6c9de5596e9fd	2026-09-01 10:00:00.000	2026-09-01 10:00:00.000	null	
2	2	s2	google	2026-09-01 11:15:00.000	8a01f26721d572b3b2b7748530a310	2026-09-01 11:15:00.000	2026-09-01 11:15:00.000	null	
3	3	s3	facebook	2026-09-02 09:30:00.000	2b89b30358ec244e1391f8b7d8e981d2	2026-09-02 09:30:00.000	2026-09-02 09:30:00.000	null	
4	4	s4	youtube	2026-09-02 14:45:00.000	e9213b647227718b9665f25a221c80	2026-09-02 14:45:00.000	2026-09-02 14:45:00.000	null	
5	5	s5	google	2026-09-03 08:20:00.000	77c8f1b2f8fb9b04c57b3ac7567d7b7	2026-09-03 08:20:00.000	2026-09-03 08:20:00.000	null	
6	6	s6	instagram	2026-09-03 16:10:00.000	e0b15444abab3159c8bf9ecf55351a3b	2026-09-03 16:10:00.000	2026-09-03 16:10:00.000	null	
7	7	s7	facebook	2026-09-04 12:00:00.000	4fb78999f9fb25d82189d5e52b414394	2026-09-04 12:00:00.000	2026-09-04 12:00:00.000	null	
8	8	s8	youtube	2026-09-04 18:30:00.000	fee0e9a3f6f794baffa10bf4541964b	2026-09-04 18:30:00.000	2026-09-04 18:30:00.000	null	

5. Add at least 2 tests to sessionId field of session_summary (2pt)

1. Show contents of models/schema.yml

models/schema.yml

```
version: 2

models:
  - name: session_summary
    description: "Analytics model for session data"
    columns:
      - name: SESSIONID
        description: "Unique identifier for each session"
        data_tests:
          - unique
          - not_null
```

dbt test

```
(data226_dbt) C:\Users\kevin\OneDrive\桌面\DATA226\HW4>dbt test
22:35:12 Running with dbt=1.12.5
22:35:13 Registered adapter: snowflake=1.12.1
22:35:13 Found 3 models, 1 snapshot, 2 data tests, 2 sources, 562 macros
22:35:13 Concurrency: 1 threads (target='dev')
22:35:13
22:35:15 1 of 2 START test not_null_session_summary_SESSIONID ..... [RUN]
22:35:17 1 of 2 PASS not_null_session_summary_SESSIONID ..... [PASS in 1.32s]
22:35:17 2 of 2 START test unique_session_summary_SESSIONID ..... [RUN]
22:35:17 2 of 2 PASS unique_session_summary_SESSIONID ..... [PASS in 0.37s]
22:35:17
22:35:17 Finished running 2 data tests in 0 hours 0 minutes and 3.95 seconds (3.95s).
22:35:17 Completed successfully
22:35:17 Done. PASS=2 WARN=0 ERROR=0 SKIP=0 NO-OP=0 REUSED=0 TOTAL=2
```

- For each step, capture screenshots accordingly (examples to follow)
- Create a github repo for this and add the link to the report

https://github.com/ching-tuhuang/data226_FAL26_HW4.git